NILAY PRAVIN MARATHE

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EDUCATION

Carnegie Mellon University **Pittsburgh, PA**

Master of Science in Mechanical Engineering – Research | GPA: 3.86 / 4.00 **May 2027**

- Coursework- Control Theory, Control Systems Integration, Deep Learning, Robot Learning, Multi-agent Path Planning

Indian Institute of Technology, Madras **Chennai, India**

Bachelor of Technology in Mechanical Engineering | GPA: 8.78 / 10.00 **July 2025**

- Coursework- Mechatronic Systems, Automation in Manufacturing, Numerical Methods, Heat Transfer

SKILLS /TECHNICAL COMPETENCIES

Programming: Proficient- Python (NumPy, Pytorch, Matplotlib, Pandas), C++, MATLAB; familiar with Simulink
Simulation: ROS 2 (Gazebo, MoveIt! and RViz); exposure to RoboDK, Webots, ABAQUS
CAD & Design: Mechanical Part Design in AutoCAD, Fusion360, Solidworks
Embedded Systems: Arduino IDE, PlatformIO, experience working with microcontrollers, especially ESP32 modules

RESEARCH PROJECTS

➢ **Carnegie Mellon University** **Pittsburgh, PA**

Multi-robot Construction Planning using Imitation Learning | ARCS Lab **January 2026 – present**

- Developed a baseline imitation learning (IL) pipeline for collective construction tasks in a simulated multi agent environment
- Generated expert demonstration datasets using a Hierarchical multi-agent collective construction planner
- Evaluated learning-based planning approaches for multi-robot planning in collaborative tasks

➢ **Indian Institute of Technology, Madras** **Chennai, India**

Multi-robot Disassembly Sequence Optimization using Genetic Algorithm | DiRO Lab **January 2025 – May 2025**

- Optimized multi-robot screw disassembly task, leading to optimal removal of 100+ screws with no human intervention.
- Designed an Electric Vehicle Battery Model Prototype to simulate disassembly sequence in RoboDK software
- Carried out simulation for 4, 5 and 6 robot workstations to find optimum value of number of robots by studying effect of number of robots on overall idle and cycle time during disassembly
- Identified crucial constraints during disassembly sequence to further reduce disassembly cycle time by 15-20%

Stochastic Methods for Path Planning of manipulators – UG Research Project **July 2024 – November 2024**

- Examined efficiency for a 4-DOF robot end effector obtained through Model predictive Path Integral (MPPI) against SIMULINK's in-built Inverse Kinematics (IK) path planner for a paintbrush end effector
- Demonstrated 2-D constrained end effector motion for PincherX100 robot arm in ROS and SIMULINK
- Reported faster performance of SIMULINK's in-built planner path by 12±2.4% as compared to the MPPI Algorithm

'AutoDock' Solution for power-assist wheelchairs – Summer Project **May 2024 – July 2024**

- Engineered a signal-strength-based algorithm in MATLAB to estimate distance between a wheelchair and a power-assist wheel's home position by using relative signal strength, and capturing wheelchair pose with a 2% error over <= 25m
- Utilized path planning algorithms A* & Dijkstra to generate optimal navigational path for power-assist wheel, enabling autonomous movement towards a wheelchair based on a scanned map of environment

COURSE PROJECTS

➢ **Carnegie Mellon University** **Pittsburgh, PA**

CitySimRL - Reinforcement Learning for Autonomous Driving in 2D Environments **January 2026 – April 2026**

- Evaluated Model-free and Mode-based Reinforcement Learning Method Performance against varying traffic density
- Designed and implemented a composite autonomous driving simulation environment by stitching multiple traffic scenarios (highway, merge, roundabout, intersection) using HighwayEnv framework by Farama Foundation
- Engineered **c**ustom lane network transitions to ensure continuous state evaluation across heterogeneous environments
- Developed realistic traffic dynamics using Intelligent Driver Model (IDM) to simulate traffic around model-run vehicle

➢ **Indian Institute of Technology, Madras** **Chennai, India**

Novel LLM Chatbot for answering queries related to Indian Philosophy **January 2025 – May 2025**

- Developed a Retrieval Augmented Generation (RAG) based LLM enabling context-aware querying of Indian Texts
- Curated a domain-specific corpus of Indian texts as an input to the Large Language Model by performing Optical Character Recognition with Gemini AI on scanned copies of old handwritten texts

Path Planning & Collision Avoidance in CNC Machines **January 2024 – May 2024**

- Implemented path-planning strategies in ROS for machining processes for CNC machine model in MoveIt! and Rviz
- Created point-cloud data of the CNC machine workspace using SLAM for collision avoidance with and without obstacles
- Modelled an interactive 3-DOF CNC station using URDF format in Fusion360, designated links, joints and coordinate frames